Millbrook Pump Station: Design Report

1 Design Basis

The Millbrook Pump Station serves a population of 21,414 persons. Its average dry weather flow is 2.09 MGD and its peak wet weather flow 12.6 MGD. The site layout is shown in Figure 4.

The wet well provides a working volume of 11,801 gal between the pump-off level and the lead pump's start level.

2 Pumps

Four submersible pumps are installed: P-301, P-302 and P-303 run together at peak wet weather flow, and P-304 is the standby.

Table 1. Pump schedule

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)	NPSHr (ft)	Speed (rpm)
P-301	1,526	71.5	106	12.5	1,379
P-302	1,397	96.2	50	16.5	1,539
P-303	2,306	81.2	85		1,205
P-304	2,169	110.9	68	9.5	TBD

P-301, the lead pump, delivers 1,526 gpm at 71.5 ft. On the standby generator at 50 Hz, P-301 delivers 1,030 gpm.

Pump P-302 has a 73 hp motor; at runout, its capacity rises to 2,908 gpm. Pump P-304's total dynamic head is 66.9 ft.

3 Wet Well

Table 2. Wet well

Working volume (gal)	12,167
Diameter (ft)	12
Depth (ft)	28.6
High water alarm elevation (ft)	427.4

The wet well's access hatch frame is low-alloy steel (ASTM A588). The access hatch is corrosion resistant in the wet well's hydrogen sulfide atmosphere.

4 Force Main and Valves

The force main is 4,224 ft of 11-inch ductile iron pipe. Surge pressures are given in Table 7.

Table 3. Valve schedule

Tag	Size (in)	Rating (psi)	Cv	Stroke time (s)
V-11	17	188	1,685	88.6
V-12	24	126	1,912	42.3
V-13	22	259	770	48.7
V-14	9		803	43.6
V-15	20	283	2,468	83.8

Table 4. Surge analysis

Case	Pressure (psi)
Maximum transient pressure	174
Minimum transient pressure	2.9

At 0.8 mi, the force main needs no intermediate air release station. The force main is laid in high-sulfate, corrosive soil along the access road.

5 Odour Control

Table 5. Odour control unit

Fan airflow (cfm)	TBD
Carbon bed volume (ft³)	167
Fan motor (hp)	3.3

6 Electrical and Materials

The pump control panel is a NEMA 1 enclosure, mounted outdoors beside the wet well.